

JEE MAIN REVISION MASTER: THERMODYNAMICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 224 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Thermodynamics - Core Application	224 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Thermodynamics - Core Application (Very High Yield — 224 PYQs)

Core Principle: Key problem pattern in Thermodynamics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity a..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): " 20.0 dm^3 of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa . Whi..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II (Expression) A. Reversible expansion I. $q=0$ B. Free expansion II. $q=nRT\ln\frac{V_2}{V_1}$ C. Irreversible Compression III. $w=-p_{\text{ext}}(V_1-V_2)$ D. Cyclic reversible IV. $\frac{q_{\text{rev}}}{T}=0$ Choose the correct answer from the options given below :

- (A) -II, B-III, C-I, D-IV
- (B) -II, B-I, C-IV, D-III
- (C) -II, B-I, C-III, D-IV **[Correct]**
- (D) -I, B-II, C-III, D-IV

Correct Answer: C

Solution Analysis: (A) Reversible isothermal expansion For an ideal gas in a reversible and isothermal expansion: $w=-nRT\ln\frac{V_2}{V_1}$ Since the process is isothermal, $\Delta U=0$ $q=-w$ Hence, $q=nRT\ln\frac{V_2}{V_1}$ Therefore, A corresponds to II. (B) Free expansion In free expansion, gas expands against zero external pressure: $p_{\text{ext}}=0$ Thus, $w=0$ and $q=0$ (Also, for an ideal gas, $\Delta U = 0$.) Therefore, B corresponds to I. (C) Irreversible isothermal compression For irreversible compression (against constant external pressure): $w=-p_{\text{ext}}(V_1-V_2)$ Therefore, C corresponds to III.